Baotong Tian

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user-tian

Education

University of Rochester **Aug, 2024 – Aug 2029 (Expected)**
Ph.D. in Electrical and Computer Engineering
GPA 4.0/4.0

Tsinghua University **Sep, 2020 – Jun, 2024**
Bachelor of Engineering in Automation
GPA 3.86/4.0

Scholarships & Awards

- **2021, 2022 Tsinghua Arts and Culture Merit Scholarship** (Awarded to undergraduate students for great contribution to cultural activities within the department)
- **2022, 2023 Tsinghua Academic Excellence Scholarship** (Awarded to undergraduate students for excelling GPA throughout the school year)
- **2023 Tsinghua Innovation Award of Science and Technology** (Awarded to undergraduate students with excellent research potential)
- **2025 WIMIR Grant from ISMIR**
- **2026 Author Grant from ISMIR**

Publications

[1] **Baotong Tian**, Zhiyao Duan. "Point-and-Play: Instrument Timbre Cloning from Music Mixtures." In submission to ICASSP 2027.

[2] **Baotong Tian**, Cynthia Lu, Vincent Cheung, Ting-Kang Wang, Jonathan Churchill, Zhiyao Duan. "VIOLET: High-Fidelity Violin Synthesis with Techhniques and Dynamics." Proc. ISMIR 2026.

[3] Yu Zhang*, **Baotong Tian***, and Zhiyao Duan. "Conan: A Chunkwise Online Network for Zero-Shot Adaptive Voice Conversion." Proc. ASRU 2025.

[4] You Zhang*, **Baotong Tian***, Lin Zhang, Zhiyao Duan. "PartialEdit: Identifying Partial Deepfakes in the Era of Neural Speech Editing." Proc. Interspeech 2025.

Research Experience

University of Rochester – *Audio Information Research Lab* **Sep, 2024 - Present**
Ph.D. Student, Advisor: Zhiyao Duan

- **Instrument Timbre Cloning from Music Mixtures** Sep, 2025 - Present
  - Proposed a novel task of cloning an instrument's timbre directly from a polyphonic mixture for synthesis of arbitrary pitches and velocities.
  - Developed **Point-and-Play**, which fine-tunes a pretrained CLAP encoder with a Cue-Gated Statistics Pooling (CGSP) module to extract the cued source's timbre without source separation.
- **Expressive Violin Performance Rendering** Jun, 2025 - Present
  - Created a paired dataset **CSV-TD** by automatically annotating expressive MIDI with note-level violin techniques and rendering corresponding audio with a commercial Kontakt sample library.
  - Proposed **VIOLET**, a framework for generating violin performances with techniques and dynamics.
- **Streaming Voice Conversion** Sep, 2024 - Present
  - Co-developed **Conan**, a chunk-wise online model for zero-shot voice conversion.
  - Adapted to model timbre and style separately, and proposed a pixel-shuffle HifiGAN for streaming applications.

- **PartialEdit: Partial Deepfake Detection** Sep, 2024 - Jun, 2025
  - Created the partially deepfake dataset **PartialEdit** using advanced neural speech-editing models.
  - Evaluated state-of-the-art utterance-level detectors and segment-level localization methods on this dataset, showing that models trained on traditional partially spoofed audio fail to detect speech edited by neural speech-editing models.
  - Discussed how artifacts introduced by codec models affect the performance of models.

Tsinghua University – *Intelligent Vision Group* **Feb, 2023 - Jun, 2024**
Undergrad Research Assistant, Advisor: Jiwen Lu & Jie Zhou

- **Unified** current approaches of Shapley value estimation.
- Proposed **SimSHAP** as a simple and fast amortized Shapley value estimator.

Work Experience

Adobe Inc. – *Speech AI Team* **May, 2026 - Present**
Research Intern (part-time since Sep. 2026), Mentor: Ke Chen, Yunyun Wang, Zeyu Jin

- Proposed an offline-to-streaming framework built on a causal diffusion transformer, adapting a pretrained offline diffusion model to generate speech chunk by chunk in real time.
- Our system enables streaming TTS and voice conversion with quality comparable to its offline teacher, while achieving competitive performance against leading baselines on both tasks.

Beijing Deepmusic Technology Co. – *R&D Department* **Jun, 2023 - Aug, 2023**
Algorithm Engineer Intern, Mentor: Jian Wu

- Applied Bi-LSTM for multi-task learning of bass and chord quality.
- Applied **Conditional Random Field** (CRF) as a post-filtering technique to enhance chord recognition task accuracy by incorporating context information.

Teaching

Teaching Assistant:

- ECE208/408 The Art of Machine Learning Spring 2025
- ECE270 Probability for Electrical Engineers Fall 2025

Professional Services

Reviewer

Journal:

- IEEE Transactions on Audio, Speech, and Language Processing (TASLP) 2025
- IEEE Signal Processing Letters (SPL) 2026
- IEEE Transactions on Multimedia (TMM) 2026

Conference:

- IEEE Automatic Speech Recognition and Understanding Workshop 2025

Volunteer

- ISMIR 2025 2025

Music Background and Interests

Singing

- Joined the school choir at primary school.
- Won the 4th place in the Singing Carnival in High School.

Instrument Playing

- Electric Piano (age 5 to 12, got the Amateur Highest Performance Level at the age of 12)
- Guitar (age 13 to present)

Music Composition & Production

- Learned basic music production skills such as music arrangement and mixing in college.
- Selected for the Class of 2021-2022 in **Tsinghua University's Music Dream Program**, which is designed to nurture and develop campus musicians.
- Released 8 songs on QQ music & The NetEase's Cloud music, produced by **Logic Pro X**.

Links

- Solo and Choir performance can be seen [here].
- My musician page on [The NetEase's Cloud Music], [QQ Music], and [SoundCloud].

Programming & Language Skills

Programming Skills:

- Proficient: Python, PyTorch, Markdown, LaTeX, Git
- Familiar: MATLAB, C/C++, Linux, TensorFlow, HTML, etc

Languages: English (Fluent), Mandarin Chinese (Native)